

OPEX Comparator

iSBR vs MBR · 25-Year Total Cost of Ownership

User Guide

Before you start: download the file to your machine first, then open it in Edge or Chrome. Do not open it directly from SharePoint or Teams. The preview window will not run the calculator.

Based on peer-reviewed industrial MBR research and the tendered Beca iSBR design data, the iSBR option presents materially lower 25-year operating cost than a comparable MBR alternative on this site.

This calculator quantifies the difference line-by-line, shows the basis for every assumption, and provides an interactive sensitivity tool for informed lifecycle decision-making.

Purpose

What this calculator does and how it is built

WHAT THE MODEL DOES

■ 25-year total cost of ownership

Capital cost plus annual operating cost (energy, labour, chemicals, sludge, maintenance, membrane replacement, downtime, unplanned) escalated and present-valued at 6.6% Fonterra corporate WACC.

■ Three scenarios

Expected case (typical Edgumbe dairy strength), IDF stress case (peak loading), and Conservative case (MBR-favourable optimised operations).

■ Bottom-up bill of materials

Chemical and engineering calculators derive consumption from Beca Process Design Report tables and IXOM supplier quotes - not allowances or rules of thumb.

■ Sourced from public-domain research

MBR-side assumptions referenced to peer-reviewed industrial MBR literature (see References slide).

■ Status and source on every parameter

Each design figure shows whether it is CONFIRMED from a primary source or a PLACEHOLDER pending data.

PEER-REVIEWED SOURCES (industrial MBR)

Click any reference for the original publication. Full list on the References slide.

[Zhang et al. \(2021\)](#) · 182-plant industrial MBR survey (China)

[Qiblawey & Judd \(2019\)](#) · Industrial immersed MBR treatability and cost

[Wei et al. \(2003\)](#) · Sludge yield in biological wastewater treatment

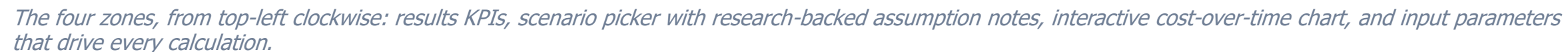
[Krampe & Krauth \(2003\)](#) · Oxygen transfer at high MLSS

[Kobayashi & Terada \(2025\)](#) · 18-year full-scale MBR case study

[IDF Bulletin 500/2019](#) · Dairy-industry body — membrane life 2-5 yr in dairy

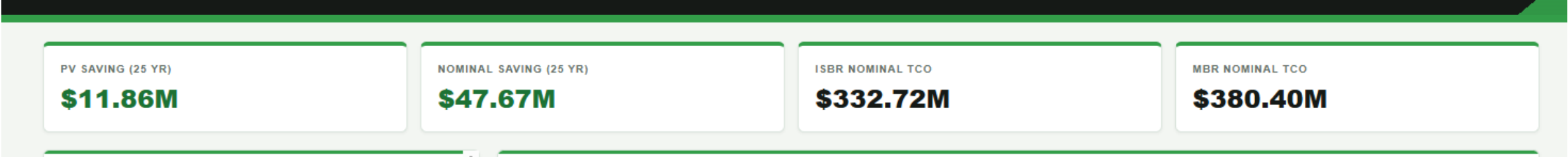
iSBR design and chemical doses: Beca Process Design Report (8303-BEC-RPT-PA-0001) Tables 14, 15, 16 and Sections 4-6. **Chemical unit prices:** IXOM supplier quote May 2026. **Discount rate:** 6.6% Fonterra corporate WACC.

Four zones - results at the top, controls on the left, charts on the right



The four results KPIs

Read these first



PV SAVING (25 YR)	NOMINAL SAVING	iSBR NOMINAL TCO	MBR NOMINAL TCO
Present value of the saving over 25 years, discounted to today at 6.6% (Fonterra corporate WACC).	Undiscounted sum of all annual cost differences over 25 years. Larger than PV because future dollars are not discounted.	iSBR 25-year total cost: CAPEX plus all escalated annual OPEX. Includes alum, polymer, RDT and centrifuge dewatering.	MBR 25-year total cost: CAPEX plus all escalated annual OPEX plus membrane replacement events plus downtime allowance.

The three scenarios

Each scenario loads a different set of assumptions

EXPECTED CASE

Typical Edgumbe dairy strength

- Energy: iSBR 19,000 / MBR 23,200 kWh/d
- Membrane life: 5 yr (Zhang 2021, 182-plant industrial survey average)
- CIP frequency: 1 maintenance clean per week, 60-day recovery clean
- Sludge: biological component at parity (Wei et al. 2003); alum +20%
- Maintenance: \$350k iSBR / \$550k MBR

Central case reflecting typical site loading. Energy from Beca Load List rev 2 plus literature uplifts for high-MLSS aeration and scour.

IDF STRESS

Peak loading, top-of-range doses

- Energy: +10% iSBR / +14% MBR for peak loading
- Membrane life: 4 yr (Zhang 2021 lower bound of 4-7 yr range)
- CIP elevated: 10 MC/week (Dupont Memcor high-fouling guide)
- Sludge at parity (Wei et al. 2003)
- Top-of-range chem doses + energy/chem creep

Sensitivity case reflecting International Dairy Federation peak loading conditions and accelerated fouling.

CONSERVATIVE

MBR-favourable optimised operations

- Energy: iSBR 17,500 / MBR 20,500 kWh/d (lower bound)
- Membrane life: 7 yr (Zhang 2021 upper bound)
- CIP relaxed: 4 MC/week + 120-day recovery clean
- Sludge at parity · No creep · No unplanned events
- Lower TP load, threshold-managed cleaning

MBR-favourable sensitivity case with optimised operations and longest defensible membrane life. Kobayashi & Terada (2025) 18-yr case shows long life is achievable but not generic.

Year-by-year schedule and lifecycle payback

The Year-by-year Schedule updates live with scenario selection - the payback year is where the Saving Cum Nominal turns from red to green

YEAR	ISBR ANNUAL	MBR ANNUAL	DELTA	ISBR CUM NOMINAL	MBR CUM NOMINAL	SAVING CUM NOMINAL
Y1	\$5.95M	\$7.14M	\$1.19M	\$130.95M	\$117.14M	\$-13.81M
Y2	\$6.11M	\$7.42M	\$1.31M	\$137.07M	\$124.56M	\$-12.51M
Y3	\$6.28M	\$7.71M	\$1.43M	\$143.34M	\$132.27M	\$-11.08M
Y4	\$6.44M	\$8.01M	\$1.57M	\$149.78M	\$140.28M	\$-9.51M
Y5	\$6.61M	\$10.62M	\$4.01M	\$156.40M	\$150.90M	\$-5.50M
Y6	\$6.79M	\$8.14M	\$1.35M	\$163.19M	\$159.04M	\$-4.15M
Y7	\$6.97M	\$8.46M	\$1.49M	\$170.16M	\$167.50M	\$-2.66M
Y8	\$7.16M	\$8.79M	\$1.64M	\$177.31M	\$176.29M	\$-1.02M
Y9	\$7.35M	\$9.13M	\$1.79M	\$184.66M	\$185.43M	\$766k
Y10	\$7.54M	\$12.09M	\$4.55M	\$192.20M	\$197.52M	\$5.32M
Y11	\$7.74M	\$9.29M	\$1.54M	\$199.94M	\$206.80M	\$6.86M
Y12	\$7.95M	\$9.65M	\$1.70M	\$207.89M	\$216.45M	\$8.56M
Y13	\$8.16M	\$10.03M	\$1.87M	\$216.05M	\$226.48M	\$10.43M
Y14	\$8.38M	\$10.42M	\$2.04M	\$224.43M	\$236.90M	\$12.47M
Y15	\$8.60M	\$13.77M	\$5.17M	\$233.03M	\$250.67M	\$17.64M
Y16	\$8.83M	\$10.59M	\$1.76M	\$241.86M	\$261.26M	\$19.40M
Y17	\$9.07M	\$11.01M	\$1.94M	\$250.93M	\$272.27M	\$21.34M
Y18	\$9.31M	\$11.44M	\$2.13M	\$260.23M	\$283.71M	\$23.48M
Y19	\$9.56M	\$11.89M	\$2.33M	\$269.79M	\$295.60M	\$25.81M
Y20	\$9.81M	\$15.68M	\$5.87M	\$279.60M	\$311.28M	\$31.68M
Y21	\$10.07M	\$12.09M	\$2.01M	\$289.67M	\$323.36M	\$33.69M
Y22	\$10.34M	\$12.56M	\$2.22M	\$300.01M	\$335.92M	\$35.91M
Y23	\$10.62M	\$13.05M	\$2.43M	\$310.63M	\$348.97M	\$38.34M
Y24	\$10.90M	\$13.56M	\$2.66M	\$321.53M	\$362.54M	\$41.00M
Y25	\$11.19M	\$17.86M	\$6.67M	\$332.72M	\$380.40M	\$47.67M

PAYBACK

Expected Case
Saving Cum Nominal
turns from red to green

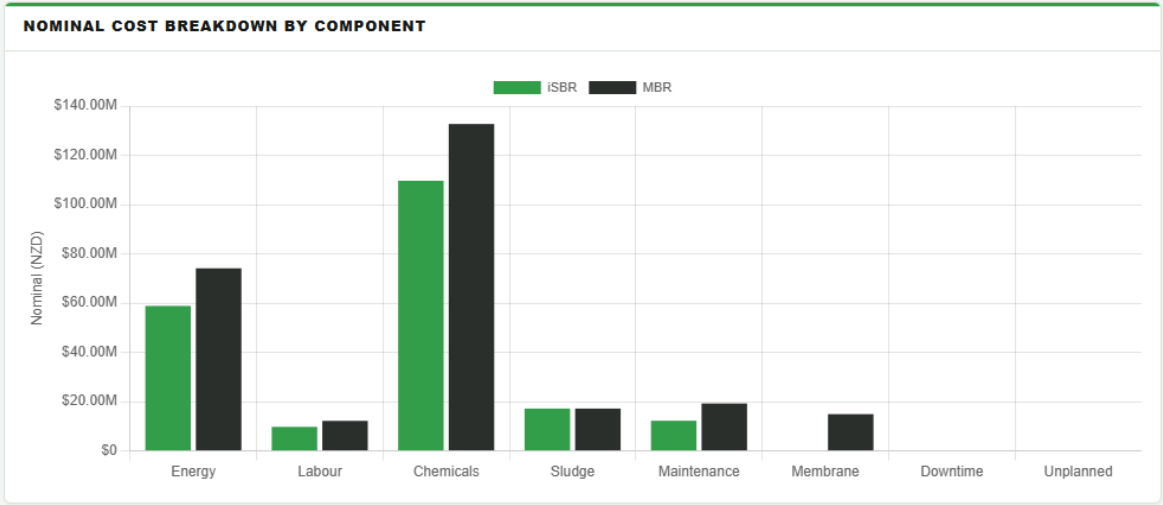
PAYBACK SHIFTS WITH SCENARIO

The live table updates when a different scenario is selected. The payback year shifts based on the operating cost gap and the CAPEX delta between options.

WHAT CHANGES PAYBACK: (1) scenario - the annual operating cost gap between iSBR and MBR; (2) CAPEX delta - the upfront cost difference between options. Editing either iSBR or MBR CAPEX in the General inputs panel moves the payback year directly.

Why MBR costs more

Waterfall bridges iSBR to MBR; component bars show the biggest lines



HOW TO READ THESE

Waterfall starts at iSBR 25-year TCO and adds each MBR cost premium line-by-line - CAPEX saves a small amount on MBR, but Energy, Chemicals and Membrane more than offset that.

Breakdown shows the same components side-by-side - useful for identifying which OPEX lines are most material.

Input parameters

Three input groups - General, Option A (iSBR), Option B (MBR)

GENERAL

■ GENERAL

Analysis period	25	years
Discount rate	6.6	% p.a.
Energy escalation	3	% p.a.
Labour escalation	3	% p.a.
General escalation	2.5	% p.a.
Electricity rate	0.226	\$/kWh
FTE fully loaded	130000	NZD/yr
Average flow	8700	m³/day
iSBR CAPEX	125000000	NZD
MBR CAPEX	110000000	NZD

OPTION A · iSBR

■ OPTION A — iSBR

Energy use	19,000	kWh/day
Yearly power cost	\$1.57M	NZD/yr
Operator FTE	2.0	FTE
Chemicals	\$3.13M	NZD/yr
Sludge disposal	\$491k	NZD/yr
Annual maintenance	\$350k	NZD/yr
Open chemical calculator (bottom-up)		
Open engineering calculator (full design basis)		
▶ EDIT CAPEX / OTHER COMMERCIAL INPUTS		

OPTION B · MBR

■ OPTION B — MBR

Energy use (Yr 1)	23,200	kWh/day
Yearly power cost (Yr 1)	\$1.91M	NZD/yr
Operator FTE	2.5	FTE
Chemicals (Yr 1)	\$3.68M	NZD/yr
Sludge disposal	\$491k	NZD/yr
Membrane life	5	years
Membrane replace cost	\$2.03M	NZD/event
Lost-production downtime per replace	0	days
Lost-production downtime cost	\$50k	NZD/day
Annual maintenance	\$550k	NZD/yr
Energy creep with age	1.5	% extra/yr
Chem creep with age	1.5	% extra/yr
Avg unplanned outage	0.0	days/yr
Unplanned outage cost	\$100k	NZD/day
Open chemical calculator (bottom-up)		
Open engineering calculator (full design basis)		
▶ EDIT CAPEX / OTHER COMMERCIAL INPUTS		

HOW IT WORKS: Pick a scenario, values load. Any cell can be overridden directly - the model recalculates live. Clicking the scenario button again reloads the defaults and resets all overrides in both bottom-up calculators.

Chemical calculator (bottom-up)

Annual chemical bill derived from design doses and supplier prices

Chemical Calculator — Option B (MBR)

Aug flow: 8,700 m³/dAnnual flow: 3,176 ML/yr

Sludge: 0.553 dry tML × 3,176 ML = 1,756 dry t/yr

Sludge = value 0.553 dry tML — industrial component of purity with OSBR (consistent with Wils et al. 2022 Water Research and Qilwasey & Judd 2020 Water Sci & Tech, which find industrial MBR observed yield similar to or lower than CASBRT due to longer SRT). Alum sludge ~20% higher than OSBR reflecting single point coagulant dosing on MBR. Not up +1.7% vs OSBR (conservative allowance — could be modified toward purity with vendor confirmation).

CHEMICAL	PURPOSE	DOSE	UNIT	UNIT COST (NZD)	ANNUAL QTY	ANNUAL COST	SOURCE / NOTES
PROCESS CHEMICALS							
Alum (48% w/w Al ₂ (SO ₄) ₃)	P removal (OSBR + media filter)	914.4	L/ML	1.82	2,963,677	\$2,96M	MBR dose 814 L/ML + OSBR baseline 762 L/ML + 1.20 single-point upflr. Basic industrial MBR P removal uses single-point coagulant dosing (no anoxic/anoxic phase separation available), which is less stoichiometrically efficient than OSBR upflr dosing — their study (2016/17) is May 2026, consistent with Qilwasey & Judd (2020) Water Sci & Tech survey of industrial MBR chemical consumption. IXOM May 2026: \$2.10/L.
Polymer (P99A, 50% emulsion)	ADT + centrifuge dewatering	45	L/50 t	6.55	79,022	\$518k	Same dewatering train as OSBR (PCT + centrifuge). Beca Table 16 basic applies identically. MBR sludge expressed at purity with OSBR per Wils et al. (2022) Water Research and Qilwasey & Judd (2020) — observed yield in MBR is similar or lower than CASBRT due to longer SRT. Polymer demand scales with sludge mass, not biodeg type.
Nitric acid (60% w/v HNO ₃)	pH down	3	L/ML	2.145	9,527	\$20k	Site-specific dose, same waste streams as OSBR — Beca Section 4 did not correct without Fonterra alkalinity data. Industrial MBR operators confirm pH adjustment is site-driven not technology-driven. Qilwasey & Judd (2020) and Zhang et al. (2022) Engineering (2022) peer-reviewed survey both note pH correction chemical demand tracks effluent alkalinity, not biodeg or process selection. Dose applied at purity with OSBR placeholder. IXOM May 2026: \$2.145/L.
Magnesium hydroxide (50% slurry)	pH up / alkalinity	15	L/ML	2.247	47,633	\$107k	Site-specific dose, same waste streams as OSBR — Beca Section 4 did not correct without Fonterra alkalinity data. Industrial MBR operators confirm pH adjustment is site-driven not technology-driven. Qilwasey & Judd (2020) and Zhang et al. (2022) Engineering (2022) peer-reviewed survey both note pH correction chemical demand tracks effluent alkalinity, not biodeg or process selection. Dose applied at purity with OSBR placeholder. IXOM May 2026: \$2.247/L.
Sodium hypochlorite (10-12.5% Cl ₂)	Re-use water disinfection	12.5	L/ML	1.395	39,694	\$55k	Same re-use distribution requirement as OSBR (Beca Table 14 basic). MBR permeate quality is typically better than OSBR effluent (post-media filter) (Zhang et al. 2022) industrial survey. Wils et al. (2016) full-scale food industry MBR, but downstream distribution demand is set by the re-use standard, not the upstream process. Dose held at purity. IXOM May 2026: \$1.395/L.
Ferrous chloride (48%) [not used]	P removal all	0	kg/ML	0.75	0	\$0	Not selected. Alum carried across to MBR for design consistency (same coagulant on both options). Ferrous is used in some industrial MBRs (Qilwasey & Judd 2020) — multiple food industry plants in survey use FeCl ₃ but switching coagulant between options would distort the comparison.
Caustic soda (NaOH) [not used]	pH up	0	kg/ML	1	0	\$0	Not selected. Sodium accumulation risk less acute in MBR (membrane decouples solids from effluent quality), but Mg(OH) ₂ retained for design consistency across both options. Industrial MBR practice mixed on this — Zhang et al. (2022) industrial survey shows both NaOH and Mg(OH) ₂ in use.
Lime [not used]	Alkalinity all	0	kg/ML	0.4	0	\$0	Not selected. Same Mg(OH) ₂ selection as OSBR for design consistency. Lime is used in industrial MBR practice — Qilwasey & Judd (2020) survey shows liquid alkalis dominate for handling reasons.
Aerifoam	Aeration foaming	0	L/ML	4	0	\$0	Not in Beca tender design. MBR foaming risk is documented — Zhang et al. (2022) Engineering and Qilwasey & Judd (2020) both note Baroscreen foaming as a recurring industrial MBR issue at high MLSS. Carried at zero placeholder, after if operators confirm need.
PROCESS chemicals subtotal						\$3.66M	
MEMBRANE CIP CHEMICALS — MBR ONLY							
Sodium hypochlorite (10% Cl ₂) [CIP]	Membrane MC + CIP cleaning	5528.75	L/yr	1.395	5,529	\$8k	Derived from CIP. 200 L/ML × scenario upflr. IXOM May 2026 price \$1.395/LC + \$2.20/LC. Dose applied as OSBR re-use water disinfection. CIP frequency basis. Zhang et al. (2022) Engineering Table 2 — 150 start industrial submerged MBR survey. MC removal range: 1.6 months (avg 4.2 months). MC removal typically weekly. Expected cases 1 MC/week + 60-day MC up concentration rate of the industrial average for daily fouling cycle. Ferrous Cl ₂ dose 2.0 kg/ML/ML reflects comparable Australian MBR operating data scaled to 3.0m dia (significantly greater). Qilwasey & Judd (2020) Water Sci & Tech industrial MBR review confirms hypochlorite as the dominant MC/PC used in food and dairy applications.

AUDIT TRAIL: Every chemical row shows the dose, unit cost, annual quantity and source. iSBR doses cite Beca Process Design Report tables (14, 15, 16). MBR doses cite peer-reviewed industrial MBR research and IXOM May 2026 supplier pricing.

DISCLAIMER: Figures are assumptions based on publicly available data and supplier quotes (refer to source notes and References slide). Where Fonterra has actual operating data we welcome the opportunity to refine these assumptions together.

Engineering calculator (design basis)

Every parameter behind the energy and chemical figures

Engineering Calculator — Option B (MBR)

Engineering basis for MBR membrane-specific chemical and energy calculations. Green rows come from design team Operational Costs also, process biological energy and process chemicals remain placeholders pending Sverdrup confirmation.

PARAMETER	VALUE	UNIT	STATUS	SOURCE / ACTION
1. MEMBRANE CONFIGURATION				
Average flow through membranes	8.7	ML/d	CONFIRMED	Design Process (Design Report 8303-001-001) (Rev 001). Design avg flow is 8.76 ML/d (used in OPEX assessment). Operational. Costs also MBR Option row 2 has 8.4 ML/d — internal data call inconsistency. Process Design Report is for authorization review deliverable to 8.7 used here. Internal review confirmed 8.4 ML/d as the design basis.
Peak flow through membranes	10.8	ML/d	CONFIRMED	Operational. Costs also MBR Option row 3.
Membrane area per module	50	m²	CONFIRMED	Operational. Costs also — typical module size.
Membrane peak flux	30	L/m²/h	CONFIRMED	Operational. Costs also MBR Option row 6.
Total required membrane area	15000	m²	CONFIRMED	Operational. Costs also — derived from peak flow / peak flux.
Number of modules	300	-	CONFIRMED	Operational. Costs also — total area / area per module.
Number of membrane trains	3	-	CONFIRMED	Operational. Costs also MBR Option row 7 + 2 trains. Expected Case scenario considers to 2 trains (alternative design input — better engineering option would require extensive 300 parameter flow 20 ML/d rather 2 trains).
Modules per train	60	-	CONFIRMED	Operational. Costs also MBR Option row 20.
Duty ratio at average flow	4	-	CONFIRMED	Operational. Costs also MBR Option row 13 — 40% redundancy.
2. MAINTENANCE CIP (MC) — HYPOCHLORITE				
Hypochlorite Cl₂ concentration	100	g Cl₂/L	CONFIRMED	Operational. Costs also MBR Option row 30. Equivalent to 100% by weight commercial types (typical 12.5%).
Hypochlorite solution density	1.2	kg/L	PLACEHOLDER	Industry typical for 12.5% Aqueous solution.
Cl₂ demand per MC event (per cell)	2	kg/MC	CONFIRMED	Operational. Costs also baseline 2.8 kg/MC/d. Expected Case scenario is 2.0 kg/MC/d (= 20 L hypochlorite) for 3 train Edgumbe based on older facility review and similar approach requiring approximately more chemical per MC (industrial MBR operating experience). Scenario (kg/m³ Cl₂/MC/MC) considers this value when loaded.
MC frequency per train	1	MC/event	CONFIRMED	Industrial MBR operating experience confirms 1 MC/event as standard practice. This matches Operational. Costs also row 30 call input (weekly). Note: row 30 scenario (78 days = 2 events = 6 MC/event) considers row 30. The row 30 weekly figure is adapted to the design basis.
Hypo solution per MC (per cell)	20.00	L/MC	DERIVED	Cl₂/kg = 1000 / Cl₂ conc per L. Operational. Costs also baseline recovers to 10 L/MC/d at 1.5 kg Cl₂. Expected Case recovers to 20 L/MC/d at 2 kg Cl₂.
Hypo MC use per week	80.00	L/week	DERIVED	daily cells = hypo per MC * MC per week. Expected Case: 4 * 20 = 80 L/week. Operational. Costs also baseline: 4 * 20 = 80 L/week using row 30 scenario interpretation, or 4 * 10 = 40 L/week using row 30 call.
Annual hypo from MC	4300.00	L/year	DERIVED	Weekly = 80. Expected Case: 80 * 52 = 4160 L/year. Operational. Costs also row 40 recovers to 3,200 L/year at 1 MC/week = 10 L/MC.
3. RECOVERY CIP (RC) — HYPOCHLORITE & CITRIC				
Days between RC	60	days	CONFIRMED	Operational. Costs also MBR Option row 34 baseline 90 days. Expected Case scenario considers to 60 days based on daily fouling profile (60 days insufficient for high organic sludge effluent). Industrial MBR operating data shows monthly RC at the upper bound of practice.
RCs per train per year	6.00	/yr	DERIVED	60 / days between RC. Operational. Costs also recovers to 4 RC at 90 days. Expected Case at 60 days = 6 RC.
Total RCs per year	18.25	/yr	DERIVED	RC per train * trains. Operational. Costs also recovers to 20.25 (5 trains = 4 RC). Expected Case 3 trains = 4.5 RC = 18.25.
Train volume during RC	30	m³	CONFIRMED	Operational. Costs also MBR Option row 52 — volume of chemicals in train during clean.
Cl₂ demand per RC per train	7.5	kg/RC/train	CONFIRMED	Operational. Costs also MBR Option row 52 — derived from train volume * cell concentration during soak.
Hypo solution per RC per train	75.00	L/RC/train	DERIVED	Cl₂/kg = 1000 / Cl₂ conc per L. Operational. Costs also row 60 recovers to 75 L/RC/train.
Annual hypo from RC	1368.75	L/year	DERIVED	total RCs * hypo per RC. Operational. Costs also row 60 recovers to 1,503 L/year at 5 trains * 60 days. Expected Case: 18.25 * 75 = 1,369 L/year.
Citric concentration in RC	10000	mg/L	CONFIRMED	Operational. Costs also MBR Option row 70.
Citric per RC per train	60	kg/RC/train	CONFIRMED	Operational. Costs also MBR Option row 77. Reconciliation: row 60 has 1,227 kg/L at 20.25 RC = 60 kg/RC. Note row 77 label says "L/RC/train" but value 60 is kg (consistency check from row 60). Industrial MBR practice confirms citric is RC only and not used in MC.

AUDIT TRAIL: Membrane configuration, maintenance CIP, recovery CIP, and chemical costs. Each row shows the value, status (CONFIRMED from primary source / PLACEHOLDER pending data) and the source citation.

DISCLAIMER: Figures are assumptions based on publicly available industrial MBR research and the tendered iSBR design data. Where Fonterra has plant-specific operating data or vendor data for the MBR alternative, McConnell Dowell would welcome the opportunity to refine the comparison.

Assumptions, limitations & invitation

Where the numbers come from and how we can refine them together

THE FIGURES IN THIS CALCULATOR ARE BASED ON

- Beca Process Design Report (8303-BEC-RPT-PA-0001) for iSBR-specific design parameters
- Peer-reviewed industrial MBR research for MBR-side assumptions (see References slide)
- IXOM supplier quote (May 2026) for chemical unit pricing
- Comparable Australian MBR operating site benchmarks for maintenance allowance and CIP frequency
- Fonterra corporate WACC of 6.6% for present-value discounting

KNOWN LIMITATIONS

- MBR-side energy is derived from the iSBR design base plus literature uplifts (high-MLSS aeration premium and scour). A vendor-specific MBR energy figure would refine this.
- Nitric acid and magnesium hydroxide doses are placeholders pending site-specific alkalinity data on each waste stream.
- Membrane vendor and scour technology are unspecified - pulse-aeration would reduce the scour energy uplift.

INVITATION

This is a first revision of the calculator and is intended to provide a transparent, defensible starting point for discussion.

Where Fonterra has:

- actual operating data from comparable plants
- site-specific alkalinity, COD or load data
- specific MBR energy or chemical data
- a preferred view on discount rate, life or escalation

McConnell Dowell would welcome the opportunity to refine these assumptions with Fonterra and present an updated lifecycle comparison.

References

Click any reference to open the original publication

1

Zhang, J., Xiao, K., Liu, Z., Gao, T., Liang, S., Huang, X. (2021) ["Large-Scale Membrane Bioreactors for Industrial Wastewater Treatment in China: Technical and Economic Features, Driving Forces, and Perspectives"](#) *Engineering* 7(6): 868-880

Used for: industrial MBR survey of 182 plants - energy intensity, membrane life range (4-7 yr), CIP intervals, sludge yield ranges, fouling indicators.

2

Qiblawey, H., Judd, S. (2019) ["Industrial effluent treatment with immersed MBRs: treatability and cost"](#) *Water Science & Technology* 80(4): 762-772

Used for: industrial MBR OPEX components, scour aeration demand (SADm), food and dairy industry case data.

3

Wei, Y., Van Houten, R.T., Borger, A.R., Eikelboom, D.H., Fan, Y. (2003) ["Minimization of excess sludge production for biological wastewater treatment"](#) *Water Research* 37(18): 4453-4467

Used for: MBR vs CAS observed sludge yield comparison - basis for biological sludge parity assumption between iSBR and MBR.

4

Krampe, J., Krauth, K. (2003) ["Oxygen transfer into activated sludge with high MLSS concentrations"](#) *Water Science & Technology* 47(11): 297-303

Used for: alpha-factor reduction at MBR-typical MLSS (10 g/L) versus iSBR MLSS (2.8 g/L) - basis for the MBR aeration energy premium.

5

Kobayashi, M., Terada, A. (2025) ["Long-term operation of a full-scale MBR for industrial wastewater treatment: a systematic evaluation during 18-year operation"](#) *Separation and Purification Technology* 355 Part A

Used for: long-term MBR performance evidence, recovery clean strategy validation - basis for Conservative case 7-year membrane life upper bound.

Project-specific sources: Beca Process Design Report 8303-BEC-RPT-PA-0001 · IXOM supplier quote May 2026 · INVENT iSBR vendor proposal · Beca Load List rev 2

References (continued)

Dairy-specific and New Zealand market sources — click any reference to open

6

Żyłka, R., Karolinczak, B., Dąbrowski, W. (2021) ["Structure and indicators of electric energy consumption in dairy wastewater treatment plant"](#) *Science of the Total Environment* 782: 146599

Used for: dairy-specific full-plant energy intensity (SCADA-measured) - validates the high plant-total energy basis for both options.

7

Żyłka, R., Dąbrowski, W., Malinowski, P., Karolinczak, B. (2020) ["Modeling of Electric Energy Consumption during Dairy Wastewater Treatment Plant Operation"](#) *Energies* 13(15): 3769 (open access)

Used for: dairy WWTP energy model - confirms DAF pre-treatment and organic load drive plant energy demand.

8

International Dairy Federation (2019) ["Wastewater Treatment in Dairy Processing"](#) - IDF Bulletin 500/2019 *International Dairy Federation, Brussels*

Used for: dairy-industry-body evidence that membrane life is 2-5 yr in dairy service (calcium fouling) - makes the 5-yr expected case the optimistic end of the dairy range.

9

Ministry of Business, Innovation & Employment (2025) ["Energy in New Zealand 2025"](#) *MBIE, Wellington, New Zealand*

Used for: New Zealand electricity pricing and industrial energy context underpinning the c/kWh rate.

Project-specific sources: Beca Process Design Report 8303-BEC-RPT-PA-0001 - IXOM supplier quote May 2026 · INVENT iSBR vendor proposal · Beca Load List rev 2